Experiment 9

AIM: Create Placement Dashboard using Data Controls & Crystal Reports

Code:

Database\

PlacementSchema.sql:

```
-- Practical 9: Placement Management Database Schema
CREATE DATABASE PlacementDb;
GO

USE PlacementDb;
GO

CREATE TABLE tbl_PlacementRecords (
    PlacementId INT IDENTITY(1,1) PRIMARY KEY,
    EnrollmentNo NVARCHAR(20) NOT NULL UNIQUE,
    StudentName NVARCHAR(80) NOT NULL,
    Department NVARCHAR(50) NOT NULL,
    CompanyName NVARCHAR(80) NOT NULL,
    JobRole NVARCHAR(80) NOT NULL,
    PackageLPA DECIMAL(5,2) NOT NULL,
    PlacementDate DATE NOT NULL,
    OfferStatus NVARCHAR(30) DEFAULT 'Placed (Selected)'
);

-- Seed Sample Records for Crystal Reports Analysis
INSERT INTO tbl_PlacementRecords (EnrollmentNo, StudentName, Department, CompanyName, JobRole,
    PackageLPA, PlacementDate)
VALUES
('92400120380', 'Anubhav Goraguddi', 'CSE', 'Microsoft India', 'Cloud Solution Architect', 28.5
    0, '2026-08-15'),
```

```
('92400120385', 'Rohit Patel', 'CSE', 'Amazon AWS', 'Software Dev Engineer', 24.00, '2026-08-22
    '),
('92400120392', 'Pooja Mehta', 'IT', 'TCS Digital', 'Full Stack .NET Developer', 9.00, '2026-09
    -01'),
('92400120401', 'Kavita Shah', 'CSE', 'Infosys Power Programmer', 'Specialist Programmer', 9.50
    , '2026-09-05'),
('92400120415', 'Deepak Joshi', 'ICT', 'Cognizant GenC Elevate', 'Backend Engineer', 6.75, '202
    6-09-12'),
('92400120422', 'Manish Trivedi', 'CSE', 'Accenture Advanced Tech', 'Associate Software Eng', 6
    .50, '2026-09-20');
```

Models\

PlacementRecord.cs:

```
using System;

namespace PlacementDashboardCrystalReports.Models
{
    public class PlacementRecord
    {
        public int PlacementId { get; set; }
        public string EnrollmentNo { get; set; } = string.Empty;
        public string StudentName { get; set; } = string.Empty;
        public string Department { get; set; } = string.Empty;
        public string CompanyName { get; set; } = string.Empty;
        public string JobRole { get; set; } = string.Empty;
        public decimal PackageLPA { get; set; }
        public DateTime PlacementDate { get; set; }
        public string OfferStatus { get; set; } = "Placed (Selected)";
    }
}
```

PlacementDashboard.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="PlacementDashboard.aspx.cs" Inherits=
    "PlacementDashboardCrystalReports.PlacementDashboard" %>

<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Placement Dashboard - ASP.NET Data Controls</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="s
    tylesheet" />
</head>
<body class="bg-light">
    <form id="form1" runat="server">
        <nav class="navbar navbar-dark bg-dark shadow-sm">
            <div class="container-fluid px-4">
                <span class="navbar-brand fw-bold">Marwadi University · Placement Cell Dashboar
    d</span>
                <span class="text-white-50 small">ASP.NET GridView &amp;amp; Data Controls (01CE152
    3)</span>
            </div>
        </nav>

        <div class="container-fluid px-4 py-4">
            <!-- Filter Controls -->
            <div class="card border-0 shadow-sm mb-4">
                <div class="card-body p-3">
                    <div class="row align-items-center">
                        <div class="col-md-3">
                            <label class="form-label small fw-bold">Filter by Department:</labe
    l>
                            <asp:DropDownList ID="ddlDepartment" runat="server" AutoPostBack="T
    rue" OnSelectedIndexChanged="ddlDepartment_SelectedIndexChanged" CssClass="form-select form
    -select-sm">
                                <asp:ListItem Text="All Departments" Value="All" Selected="True
```

```
" />
                        <asp:ListItem Text="Computer Science &amp;amp; Engineering" Value="
CSE" />
                        <asp:ListItem Text="Information Technology" Value="IT" />
                        <asp:ListItem Text="Information &amp;amp; Comm. Tech" Value="ICT" /
>
                    </asp:DropDownList>
                </div>
                <div class="col-md-9 text-end pt-3">
                    <asp:Button ID="btnGenerateReport" runat="server" Text="Open Crysta
l Reports Viewer" OnClick="btnGenerateReport_Click" CssClass="btn btn-danger btn-sm px-3 sh
adow-sm" />
                </div>
            </div>
        </div>
    </div>

    <!-- ASP.NET GridView Control -->
    <div class="card border-0 shadow-sm">
        <div class="card-header bg-white py-3">
            <h6 class="mb-0 fw-bold">Student Placement Records GridView</h6>
        </div>
        <div class="card-body p-0">
            <asp:GridView ID="gvPlacementRecords" runat="server" AutoGenerateColumns="F
alse"
                          CssClass="table table-hover align-middle mb-0" GridLines="Non
e"
                          AllowPaging="True" PageSize="10" OnPageIndexChanging="gvPlace
mentRecords_PageIndexChanging">
                <Columns>
                    <asp:BoundField DataField="EnrollmentNo" HeaderText="Enrollment No"
 ItemStyle-Font-Bold="True" />
                    <asp:BoundField DataField="StudentName" HeaderText="Student Name" /
```

```
    >
                        <asp:BoundField DataField="Department" HeaderText="Department" />
                        <asp:BoundField DataField="CompanyName" HeaderText="Company" />
                        <asp:BoundField DataField="JobRole" HeaderText="Job Profile" />
                        <asp:BoundField DataField="PackageLPA" HeaderText="Package (LPA)" D
    ataFormatString="{0:F2} LPA" />
                        <asp:BoundField DataField="PlacementDate" HeaderText="Offer Date" D
    ataFormatString="{0:dd-MMM-yyyy}" />
                        <asp:BoundField DataField="OfferStatus" HeaderText="Status" />
                    </Columns>
                </asp:GridView>
            </div>
        </div>
    </div>
</form>
</body>
</html>
```

PlacementDashboard.aspx.cs:

```
using System;
using System.Data;
using System.Data.SqlClient;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace PlacementDashboardCrystalReports
{
    public partial class PlacementDashboard : Page
    {
        private readonly string connStr = "Server=.;Database=PlacementDb;Trusted_Connection=Tru
    e;";

        protected void Page_Load(object sender, EventArgs e)
```

```
    {
        if (!IsPostBack)
        {
            BindPlacementGrid();
        }
    }

    private void BindPlacementGrid()
    {
        using (SqlConnection con = new SqlConnection(connStr))
        {
            string query = "SELECT EnrollmentNo, StudentName, Department, CompanyName, JobR
ole, PackageLPA, PlacementDate, OfferStatus FROM tbl_PlacementRecords";
            if (ddlDepartment.SelectedValue != "All")
            {
                query += " WHERE Department = @Dept";
            }
            query += " ORDER BY PackageLPA DESC";

            using (SqlCommand cmd = new SqlCommand(query, con))
            {
                if (ddlDepartment.SelectedValue != "All")
                {
                    cmd.Parameters.AddWithValue("@Dept", ddlDepartment.SelectedValue);
                }
                using (SqlDataAdapter da = new SqlDataAdapter(cmd))
                {
                    DataTable dt = new DataTable();
                    da.Fill(dt);
                    gvPlacementRecords.DataSource = dt;
                    gvPlacementRecords.DataBind();
                }
            }
```

```
            }
        }

        protected void ddlDepartment_SelectedIndexChanged(object sender, EventArgs e)
        {
            BindPlacementGrid();
        }

        protected void gvPlacementRecords_PageIndexChanging(object sender, GridViewPageEventArg
    s e)
        {
            gvPlacementRecords.PageIndex = e.NewPageIndex;
            BindPlacementGrid();
        }

        protected void btnGenerateReport_Click(object sender, EventArgs e)
        {
            Response.Redirect("PlacementReportViewer.aspx");
        }
    }
}
```

PlacementReportViewer.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="PlacementReportViewer.aspx.cs" Inheri
    ts="PlacementDashboardCrystalReports.PlacementReportViewer" %>
<%@ Register Assembly="CrystalDecisions.Web, Version=13.0.3500.0, Culture=neutral, PublicKeyTok
    en=692fbea5521e1304" Namespace="CrystalDecisions.Web" TagPrefix="CR" %>

<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Crystal Reports Web Viewer - Placement Report</title>
</head>
<body>
```

```
    <form id="form1" runat="server">
        <div>
            <CR:CrystalReportViewer ID="CrystalReportViewer1" runat="server"
                                    AutoDataBind="true"
                                    ToolPanelView="None"
                                    HasExportButton="True"
                                    HasPrintButton="True"
                                    HasZoomFactorList="True" />
        </div>
    </form>
</body>
</html>
```

PlacementReportViewer.aspx.cs:

```
using System;
using System.Data;
using System.Data.SqlClient;
using CrystalDecisions.CrystalReports.Engine;

namespace PlacementDashboardCrystalReports
{
    public partial class PlacementReportViewer : System.Web.UI.Page
    {
        private ReportDocument rptDoc;

        protected void Page_Init(object sender, EventArgs e)
        {
            ConfigureCrystalReport();
        }

        private void ConfigureCrystalReport()
        {
            rptDoc = new ReportDocument();
```

```
            string rptPath = Server.MapPath("~/Reports/PlacementReport.rpt");
            rptDoc.Load(rptPath);

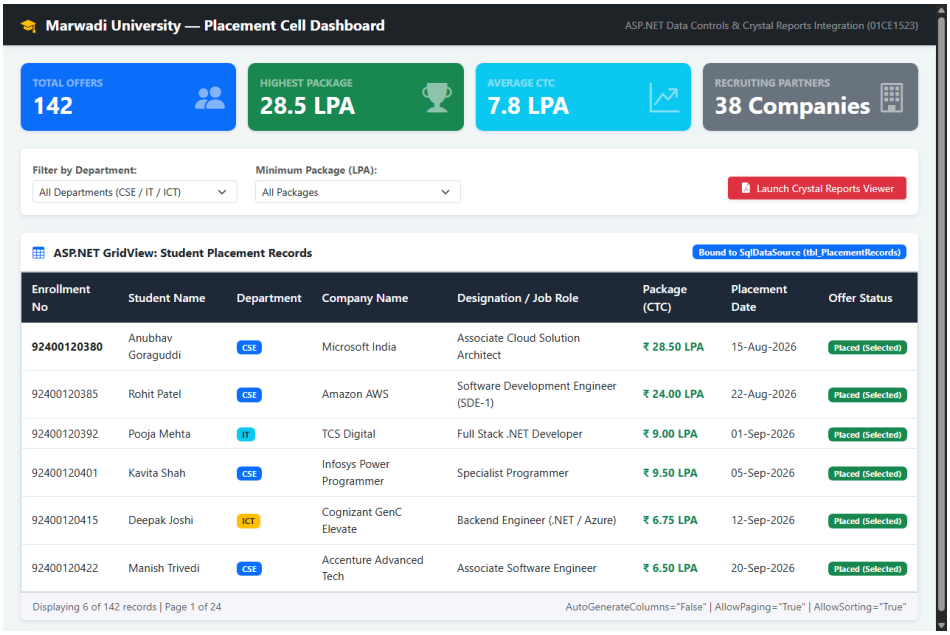
            string connStr = "Server=.;Database=PlacementDb;Trusted_Connection=True;";
            using (SqlConnection con = new SqlConnection(connStr))
            {
                string sql = "SELECT * FROM tbl_PlacementRecords ORDER BY Department, PackageLP
    A DESC";
                using (SqlDataAdapter da = new SqlDataAdapter(sql, con))
                {
                    DataTable dt = new DataTable("tbl_PlacementRecords");
                    da.Fill(dt);
                    rptDoc.SetDataSource(dt);
                }
            }

            CrystalReportViewer1.ReportSource = rptDoc;
        }

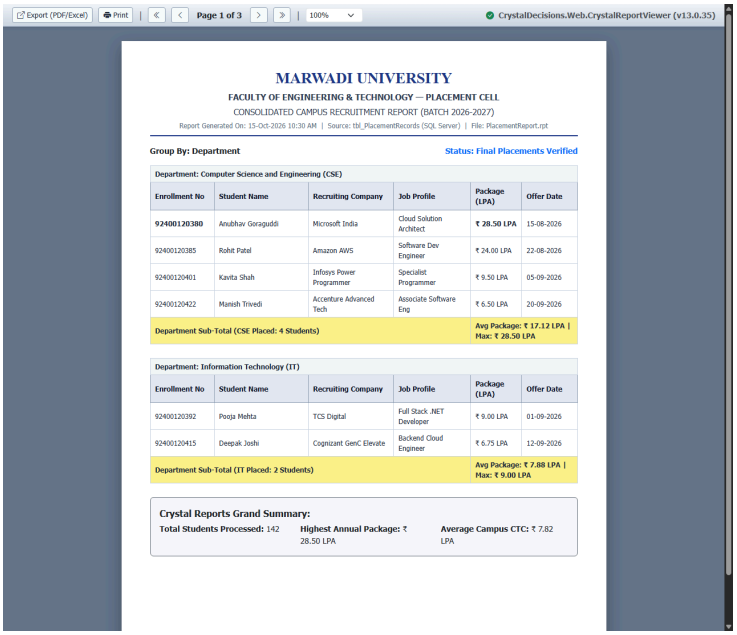
        protected void Page_Unload(object sender, EventArgs e)
        {
            if (rptDoc != null)
            {
                rptDoc.Close();
                rptDoc.Dispose();
            }
        }
    }
}
```

Output: Screenshot:

http://localhost:5001/PlacementDashboard.aspx (Data Controls & GridView)



http://localhost:5001/PlacementReportViewer.aspx (Crystal Reports Web Viewer)



GitHub URL: https://github.com/Anubhavg1123/Practical-9